Blockchain Explained

Lecture 3-4

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UNDERSTANDING BARTER SYSTEM EXCHANGE LOGIC

START: Ox and Multi-point Needs

The farmer has a



1. EXCHANGING OX FOR INTERMEDIARY GOOD (WHEAT bags)

The farmer finds an intermediary who takes the non-divisible ox and provides transferable, divisible bags of wheat.



2 EARLY COMMODITIES USED


Cowrie Shells


Salt bags


Tea leaves


Tobacco leaves


Cattle (Cow/Goat)


Seeds

Before coins notes, people these vari commodities form of "mo to facilitate t

3 HOW DID THINGS GET EXCHANGED?



Farmer has a whole variety of needs but only an ox to spare.

CHALLENGES

*DOUBLE COINCIDENCE OF WANTS



Finding someone who has exactly what you need AND wants exactly what you have.

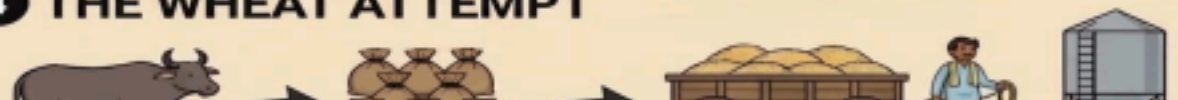
**LACK OF COMM STANDARD OF VA

Comparing the value o good against another (e.g., 1 ox vs. 1 pair of sh

**TRUST & OWNERSHIP ISSUE



4 THE WHEAT ATTEMPT





Stone Money of

1000

Years Ago


SIKKIM Bhutan
ASSAM
NAGALAND
MEGHALAYA
MANIPUR
Bangladesh
TRIPURA
BENGAL
MIZORAM
Myanmar
(Burma)
GUIZHOU
HUNAN
JIANGXI
FUJIAN
YUNNAN
GUANGXI
GUANGDONG
Taiwan
Laos
HAINAN
Philippine Sea
Paracel
Islands
South
China Sea
Luzon
Thailand
Vietnam
Cambodia
Philippines
Panay
Andaman Sea
Spratly
Islands
Gulf of
Thailand
Palawan
Negros
Sulu Sea
Mindanao
Palau



HOW YAPESE TRIBE USED BLOCKCHAIN BEFORE THE WORLD KNEW

A brilliant system of trust, memory, and transparency—thousands of years before technology.

1 THE PROBLEM: IMMOVABLE CURRENCY



The Asset:
The Yapese people used large, circular rocks called Rai Stones as currency.

The Challenge:
Each stone weighed around 200 kilos and was up to 2 meters tall.

The Result:
Physically exchanging a stone for goods or services was far too difficult.

2 THE SOLUTION: FRACTIONAL & MENTAL OWNERSHIP

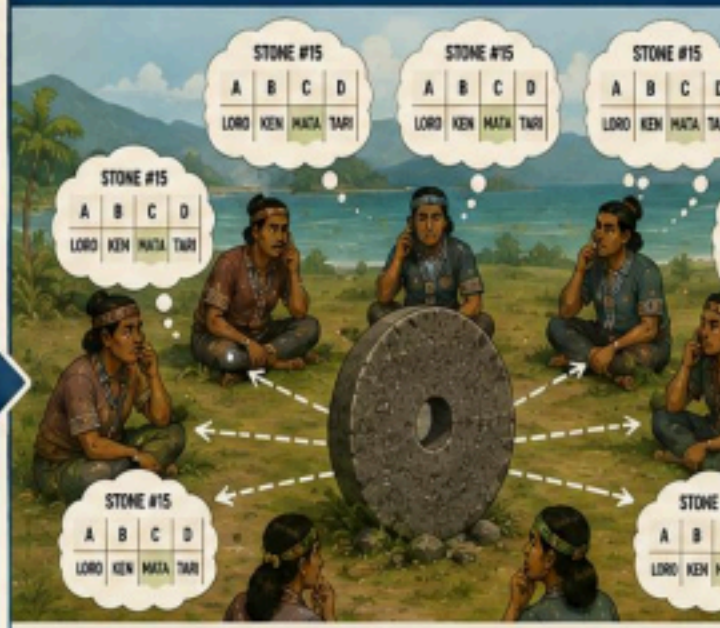


Dividing Value:
Instead of moving the stones, the tribe divided them into conceptual "sections."

Public Broadcasting:
When a trade occurred, the transacting parties stood before the tribe and publicly announced the exchange.

Mental Bookkeeping:
Every adult in the tribe memorized who owned which specific part of each stone.

3 THE MECHANISM: A DISTRIBUTED LEDG



Continuous Updates:
Every time a new transaction was announced, every tribe member updated their own "mental ledger."

Total Transparency:
All Rai stone ownership was known to everyone in the community simultaneously.

A True Di
No single
the ledge
Trust was
memory a

How did the Yapese manage the ledger?

Distributed Ledger

All tribe members keep a copy of the ledger in their head. Everyone knew who owned which Rai stone at any time

Share transaction

When two parties wished to transact, they would announce their transaction to the tribe



Update Ledger

When a transaction was announced, all tribe members updated their mental ledger

.

THE EVOLUTION OF MONEY: FROM BARTER TO DIGITAL CURRENCY

1. BARTER SYSTEM (ANCIENT TIMES)



Direct exchange of goods and services without a medium of exchange.

2. COMMODITY MONEY (c. 9000-6000 BCE)





Use of items with intrinsic value as currency, such as Cowrie Shells, Salt Blocks, Cattle, and Rai Stones.

3. METALLIC MONEY MONEY & COINAGE (c. 700 BCE)





Standardized precious metal coins (gold, silver, copper).

4. PAPER MONEY (c. 7th Century CE - 19th Century)





Representative money backed by reserves (initially gold or silver).

5. MODERN ELECTRONIC MONEY (Late 20th Century - Present)





Digital transactions, credit cards, online banking.

6. DIGITAL CURRENCY & CRYPTOCURRENCY (21st Century)

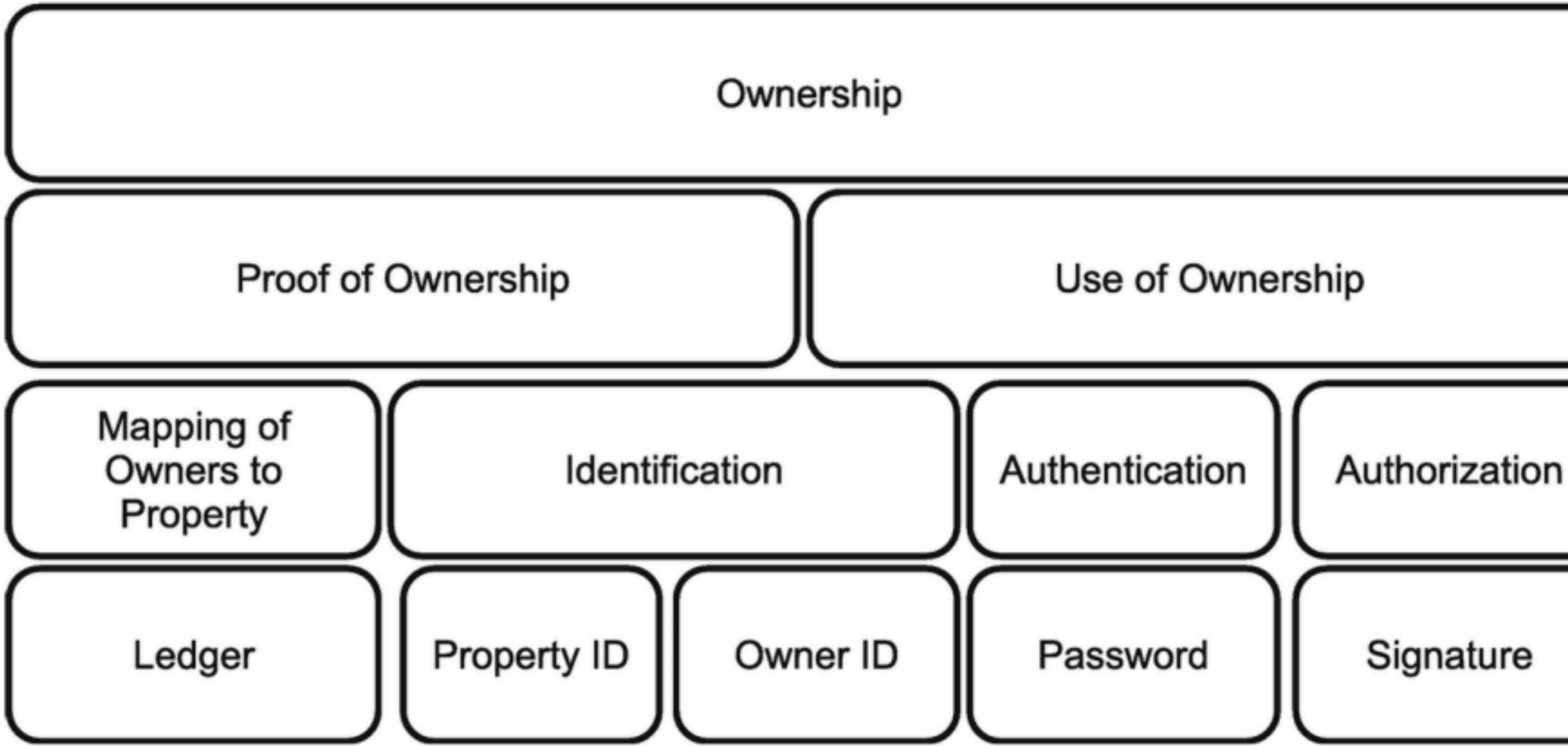




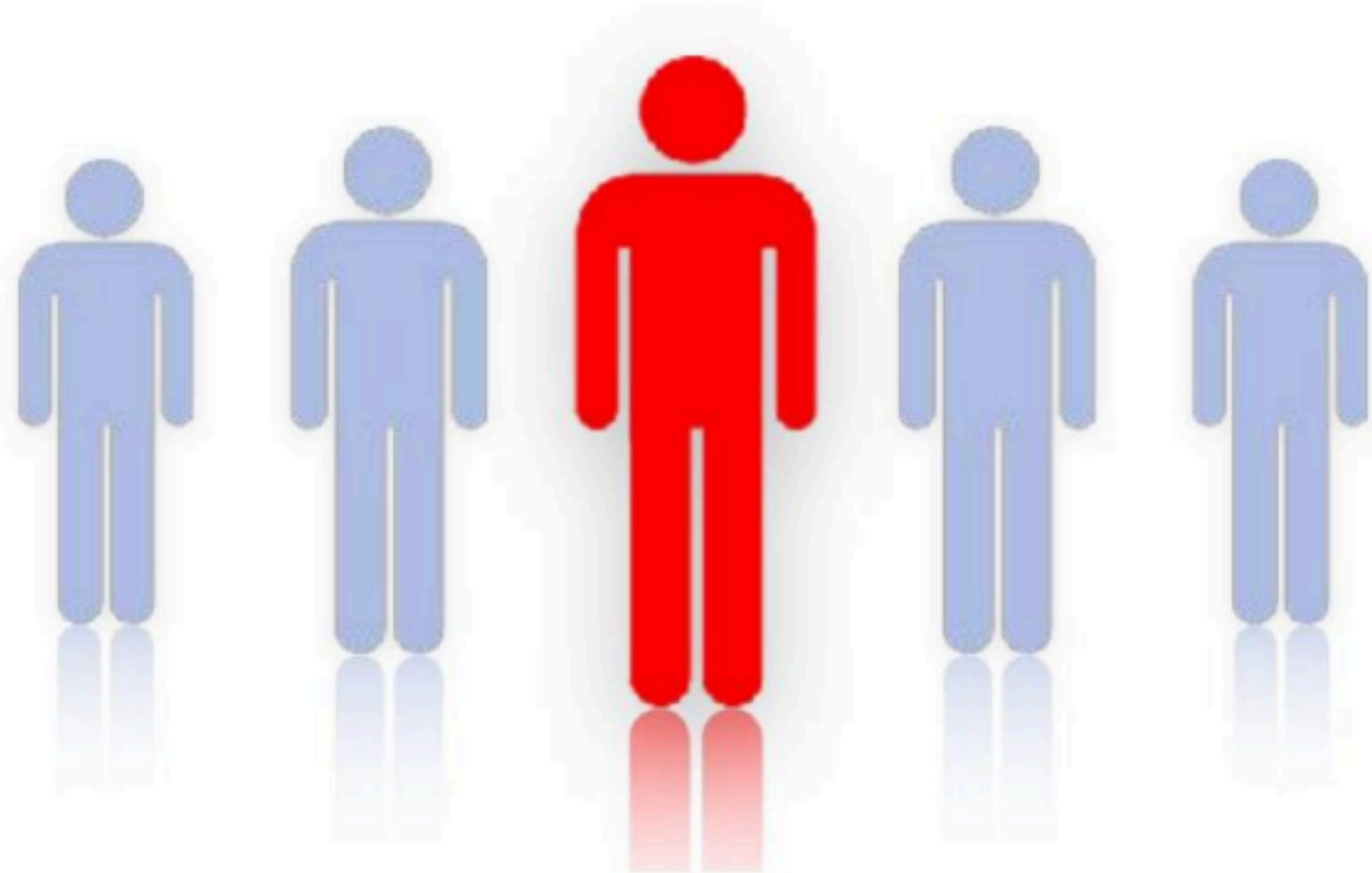
electronic currency based on cryptography.

The Internet of Information



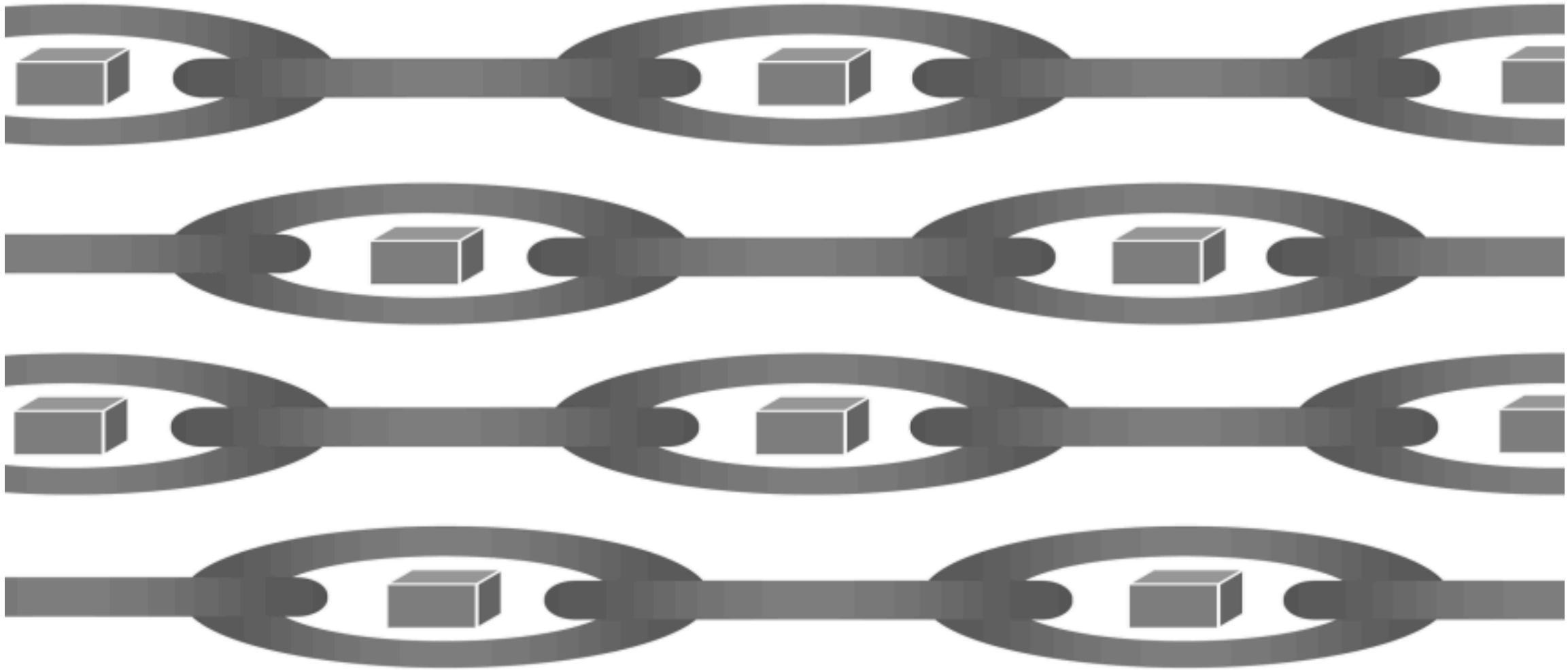

Ownership
Proof of Ownership
Use of Ownership
Mapping of Owners to Property
Identification
Authentication
Authorization
Ledger
Property ID
Owner ID
Password
Signature

Need for Middle Man: The trusted third



party

Blockchains allow users to easily prove their ***identities***, protect ***ownership*** of digital assets, and ***verify transactions*** without a high-cost intermediary



The Term is ambiguous

Blockchain has different meanings for different people depending on the context.

The term blockchain can refer to:

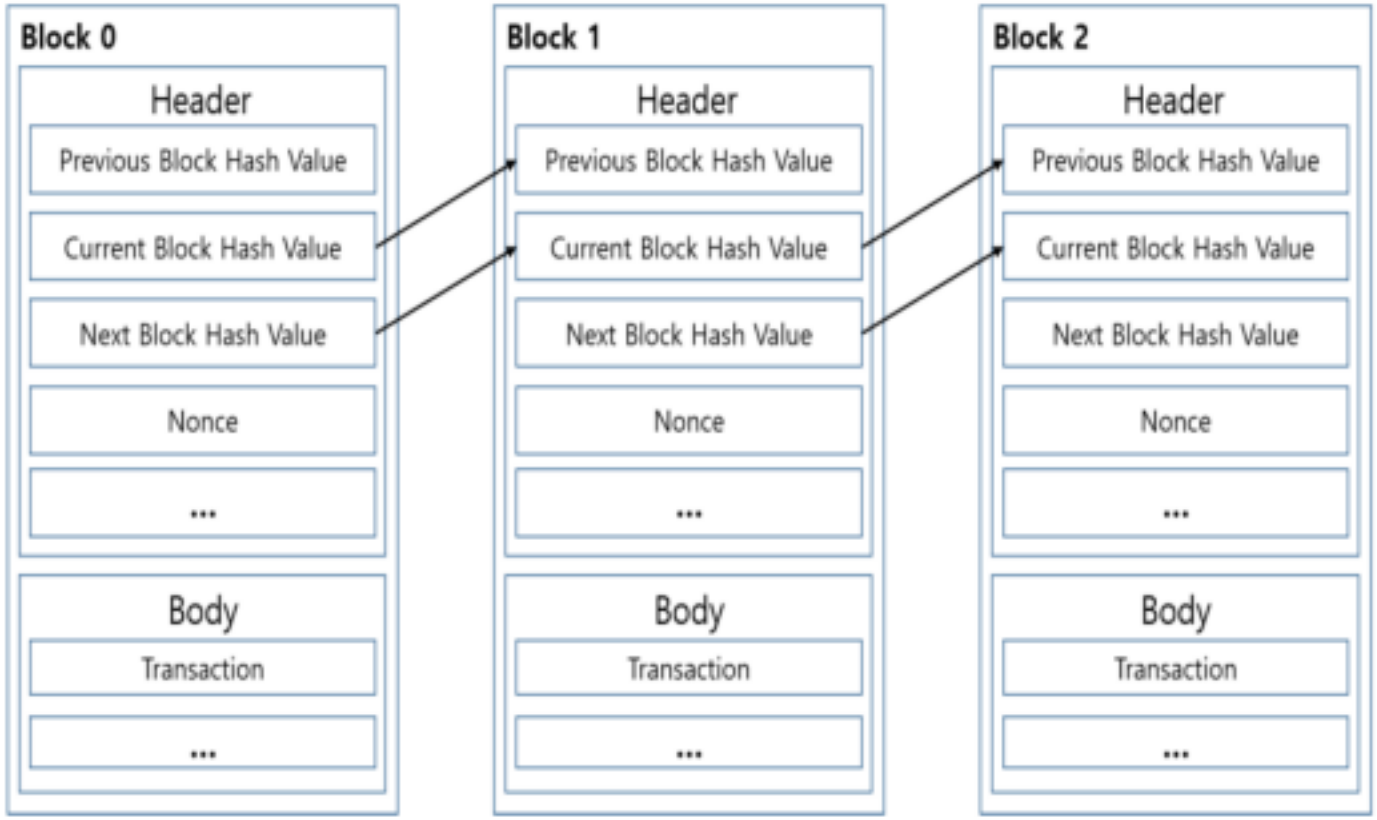
- The blockchain data structure.
- The blockchain algorithm.
- The blockchain technology suit.
- A purely distributed peer-to-peer systems of ledgers.

However, the accumulated definition of Blockchain is :

• *a distributed peer-to-peer system of ledgers that utilizes an algorithm*

or • a data structure that functions as a distributed, immutable ledger, •

which negotiates the informational content of ordered and connected blocks of data to record transactions.



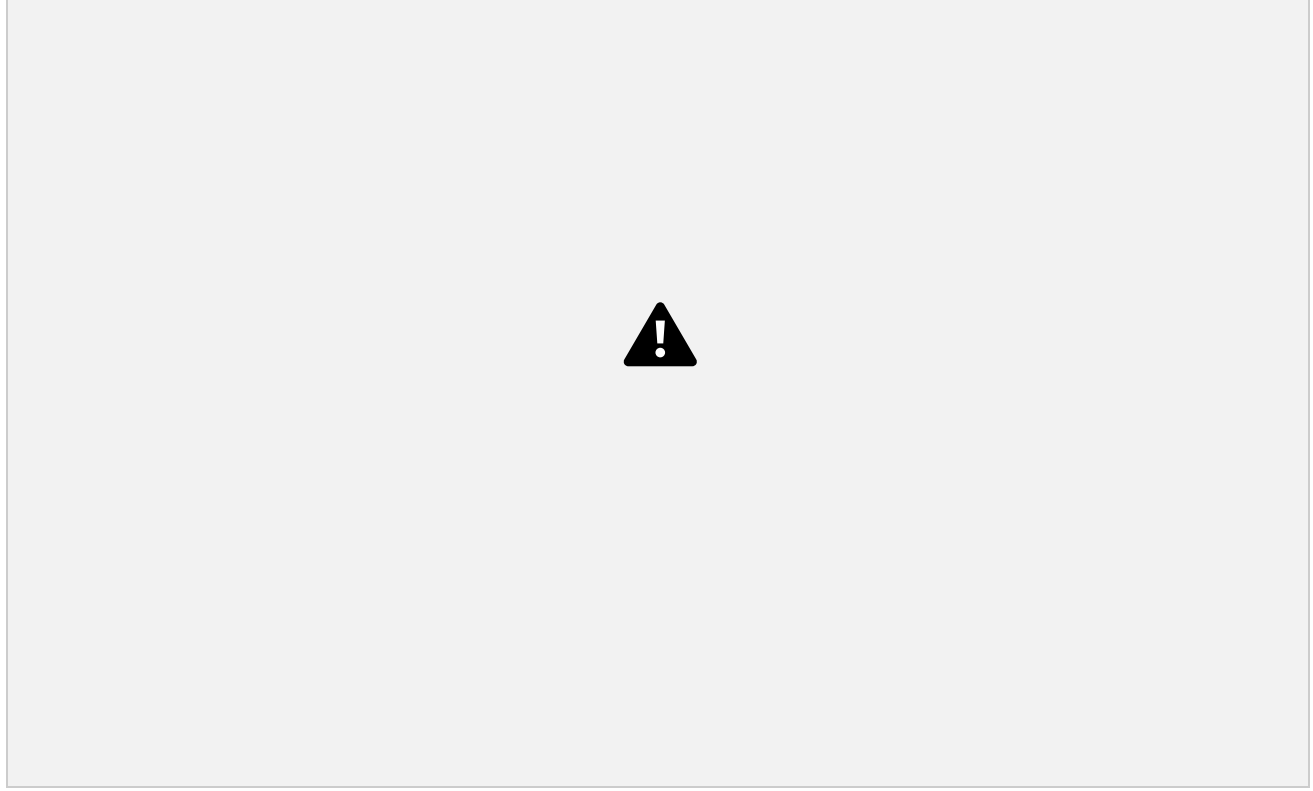
Hashing

Linked List Vs Blockchain

Feature	Linked List	Blockchain
Structure	Nodes connected by pointers/references	Blocks connected using cryptographic hashes
Linking Mechanism	Memory addresses (next pointer)	Hash of the previous block

Data Integrity	No built-in protection against modification	Any modification changes the hash, making tampering detectable Replicated across many computers (nodes)
Storage Location	Usually stored in a single computer's memory	Shared and synchronized among many participants
Ownership	Controlled by one application or system	Previous blocks are effectively immutable once confirmed
Modification	Nodes can be easily inserted, deleted, or updated	
Verification	No verification mechanism	Every node verifies new transactions/blocks
Failure	Single system failure can lose the list	Highly fault tolerant due to distributed copies

Blockchain Network



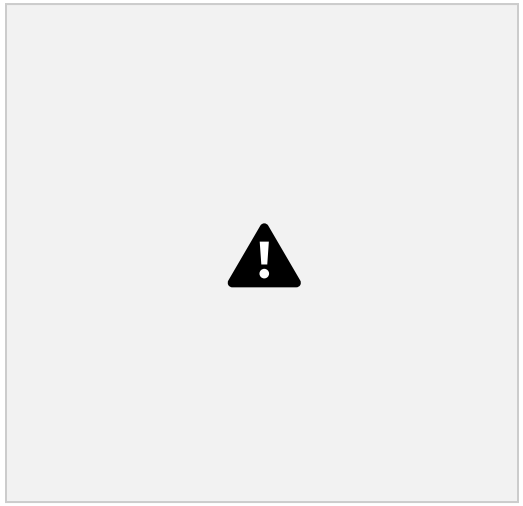
Blockchain network refers to a collection of computing nodes that holds a copy (full or partial) of the blockchain.

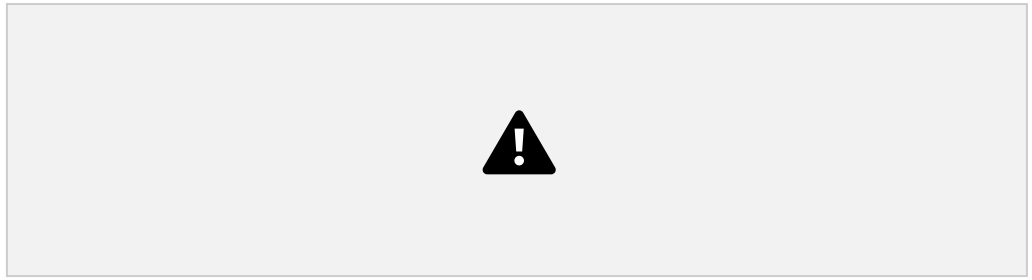
Blockchain Properties

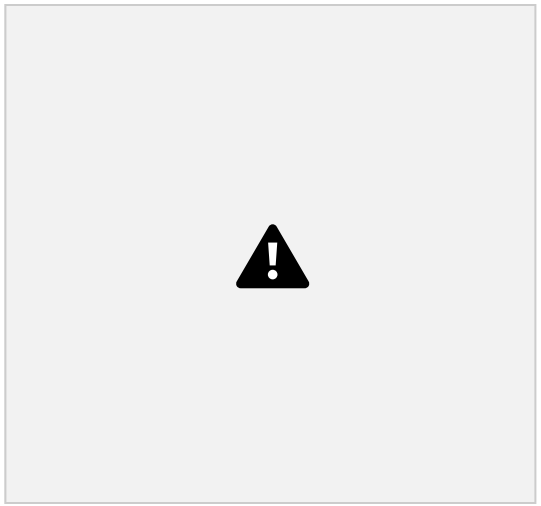
- Immutability

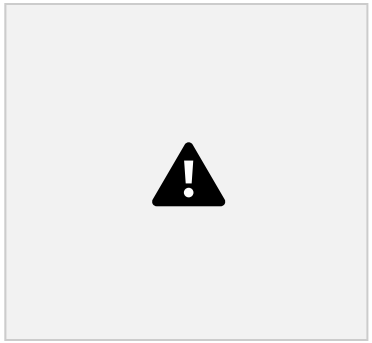
- Consensus
- Fairness of transaction order
- Decentralization
- Fault Tolerance
- Transparency

Immutability











in practical terms:

" The maximum degree of difficulty to change something "

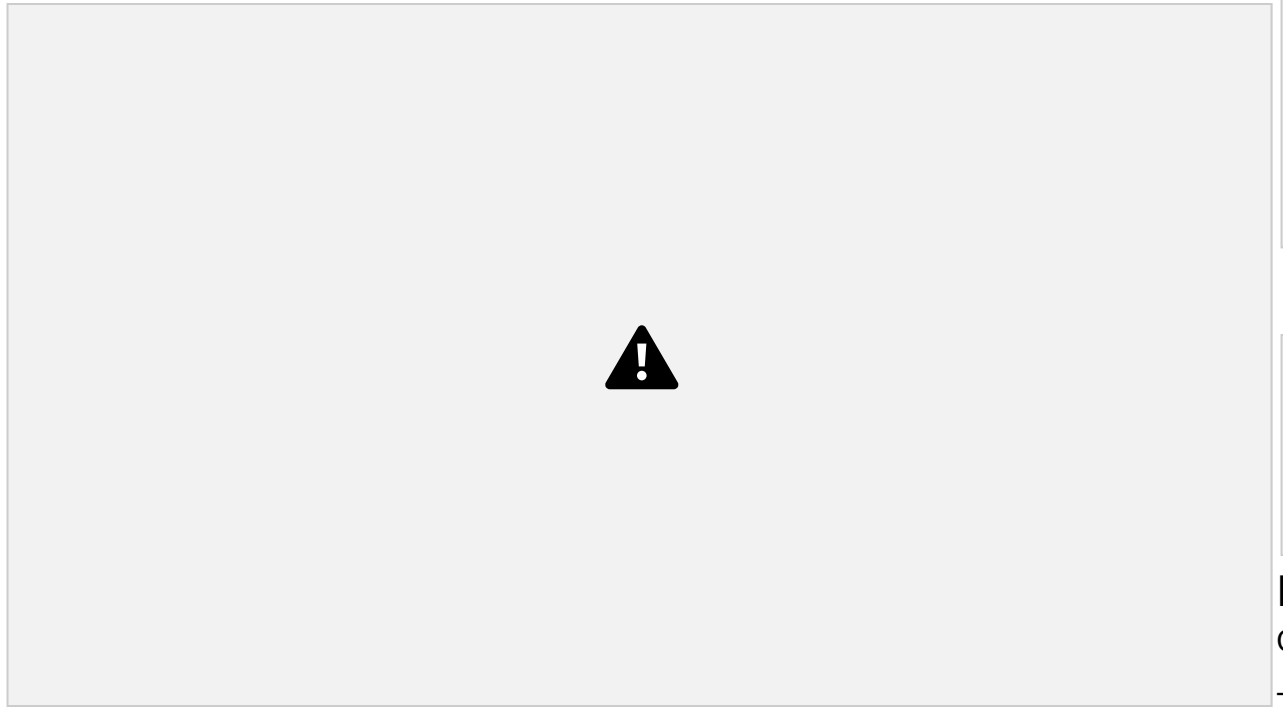


Consensus

All the parties of the blockchain network come to a common agreement (consensus) on the present state of the ledger and be able to trust unknown peers in a distributed computing environment.

Consensus Protocol

All the parties of the blockchain network come to a common agreement (consensus) on the present state of the ledger and be able to trust unknown peers in a distributed computing environment.







Proof of Work

Currently the most common and one of the most robust consensus mechanism.

The network validator has to solve
mathematically complex puzzles on the new block before approving the block to the ledger



Proof of Stake

During the validation of new blocks on the blockchain network. PoS differs with the PoW mining consensus with a mechanism where blocks are validated based on the stake of the network participants.

A consensus decision is made based on the total number of decisions submitted by all the generals. This approach forces a low overhead on the performance of the replicated service.

Practical Byzantine Fault Tolerance



Blockchain Transaction

In a blockchain based application, information is exchanged via communicating entities in the form of a transaction. A transaction has four main parts:

- ❏ a sender
- ❏ a recipient list
- ❏ the content that is being transferred
- ❏ a signed hash of all preceding transaction fields
- ❏ For each new transaction or a set of transactions, a new block is added to the blockchain.



Sender



Having no central authority, based on peer-to-peer networks with consensus algorithms.

Is a distributed ledger used to maintain a continuouslygrowing list of records called blocks.

Computer protocols that facilitate, verify, or

enforce the negotiation or performance of a contract.

What Makes Blockchain Secure

Cryptographic Hashing:
Uses mathematical algorithms like SHA-256 to turn data into unique codes. Changing even one tiny detail breaks the whole chain.

Decentralization:
Spreads data across many computers (nodes) so there is no single point of failure.

Consensus Mechanisms:
Requires network agreement (like Proof of Work or Proof of Stake) before adding new blocks

TYPES OF BLOCKCHAIN

Hybrid Blockchain

A hybrid blockchain is a combination of the private and public blockchain. Only a selected section of data or records from the blockchain can be allowed to go public keeping the rest as confidential in the private network

Consortium Blockchain

More than one organization take part in maintaining the network . However, these are mainly permissioned.







Public Blockchain

Here no one is in charge and anyone can participate in the blockchain. Anyone having access to the internet can become an authorized node by signing into the

blockchain network

Private Blockchain

This is permissioned network that impose authentications and restrictions on who is allowed to participate in the network and in what transactions.

Write down FOUR key advantage of blockchain over traditional system

Blockchain Advantages

- No Middlemen (Disintermediation)

• Immutability (Unchangeable data) •

Ensure Trust & Authenticity •

Secure Transaction

• Transparency

• Better Speed

• Smart Contracts Automation

Why/Where You Need Blockchain

• Multiple parties share data

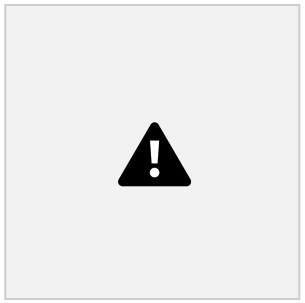
- You do not fully know, trust, or want to rely on one single company
- You still need a third-party/ middle man
- You have trust issue
- You want work to be done faster

List FIVE use case where you should apply Blockchain technology



Distributor

Central Warehouse

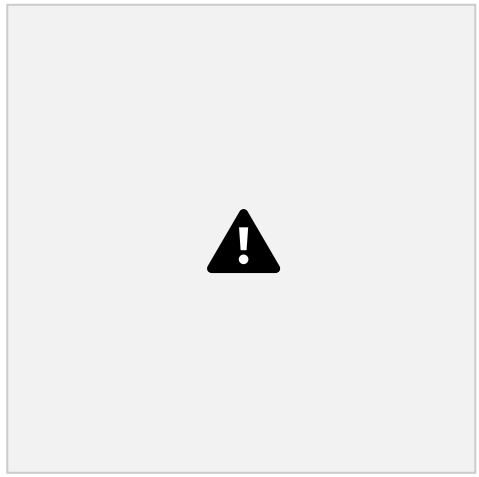


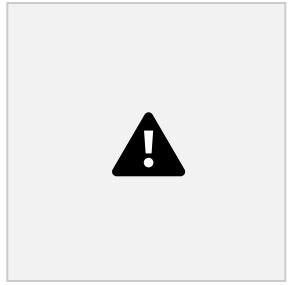




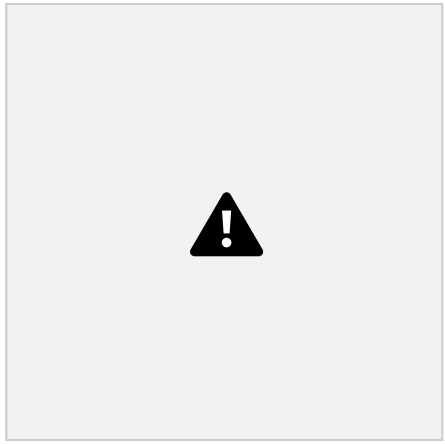
Manufacturer

Retailer





Custom



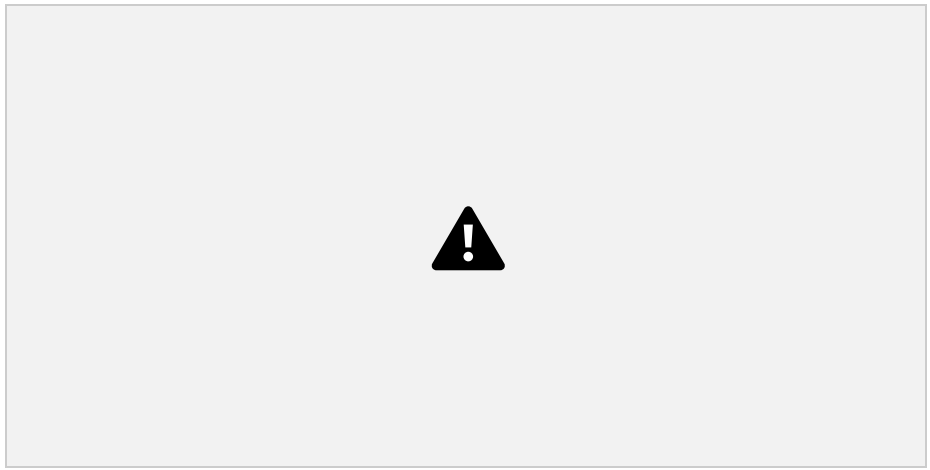
er



Wholesaler

Shopowner

What is the problem ?





Central Warehouse

Distributor

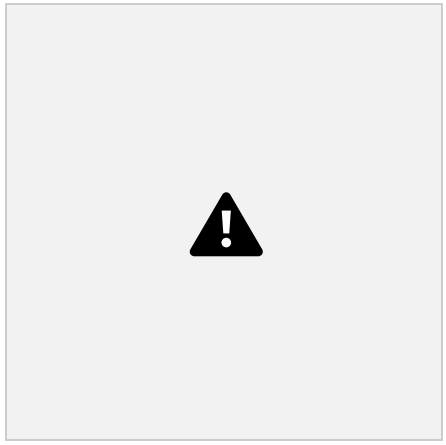
Retailer

Customer



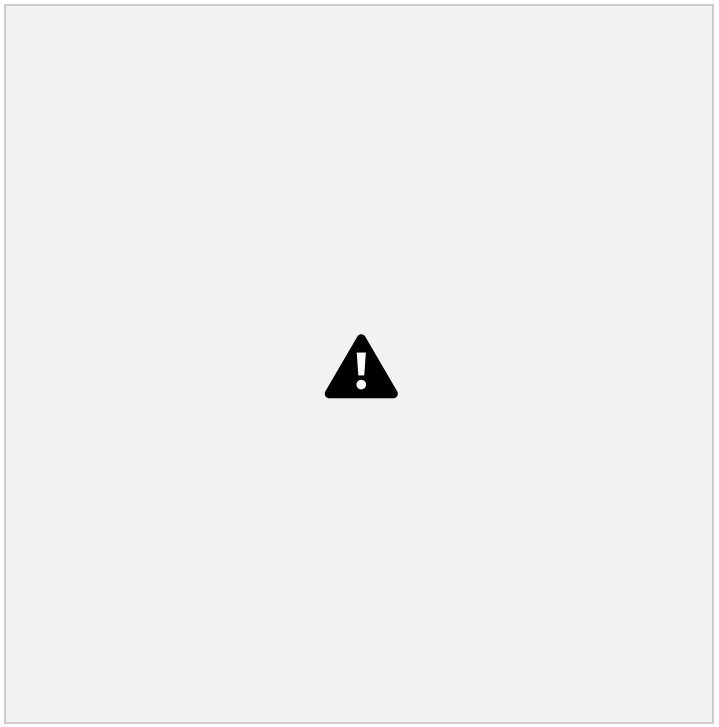


Wholesaler



Shop Owner

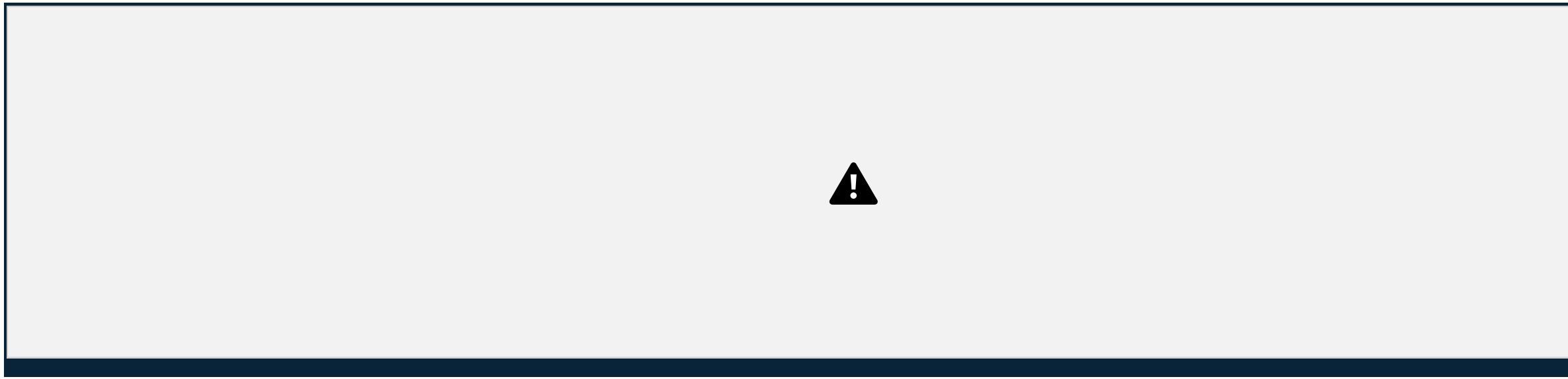








BLOCKCHAIN IN SCM



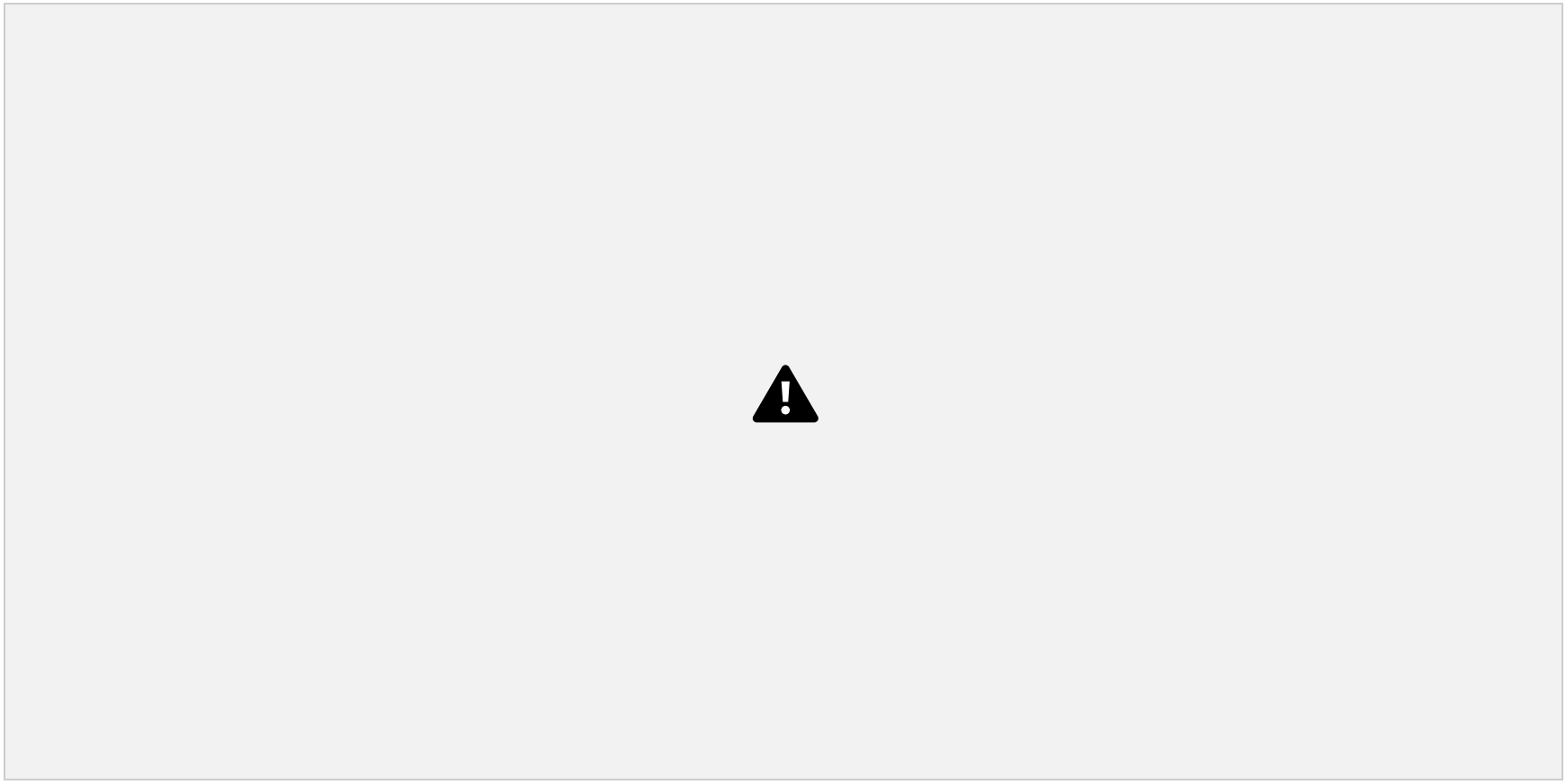
List some Identity Stage for Human Lifecycle

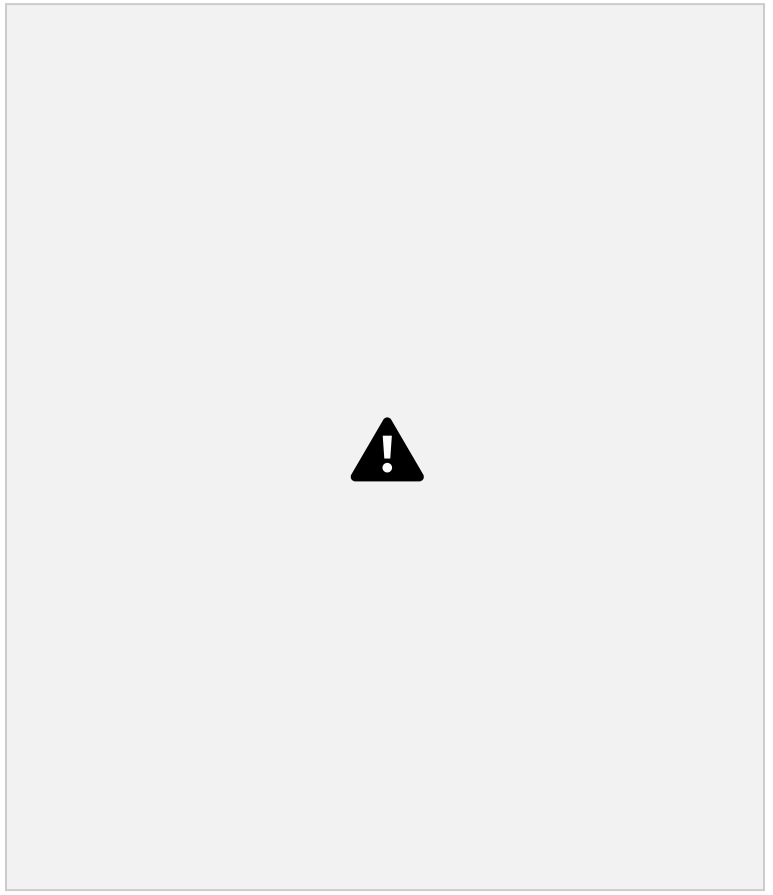
BLOCKCHAIN IN





Blockchain in Health





Blockchain in Education



Blockchain based online review

Consumers love using online reviews and they are heavily influenced by reviews

Yet many of them…..
Don't trust online reviews or ratings
Don’t trust review content

So…Ensuring immutability to produce



ossible transparency by blockchain

Thank You